Edge AI and Machine Learning

Romina Soledad Molina, Ph.D.

Ciudad Real, Castilla-La Mancha, España - 2026

KaleidoForge
Lightning Bridges for Fast AI

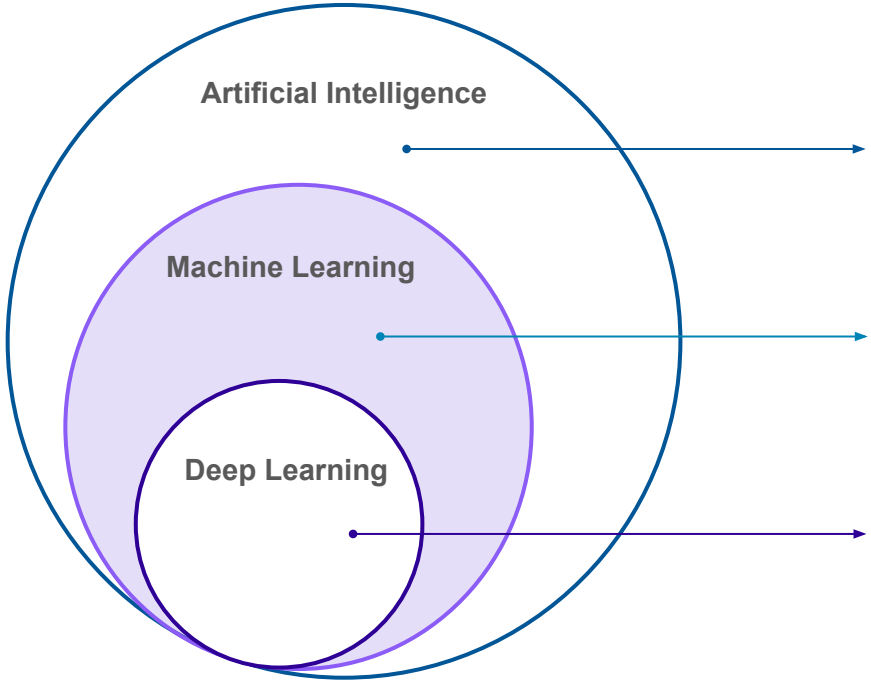
UCLM
Universidad
Castilla-La Mancha

Outline

- Introduction to Machine Learning and Edge AI
- Edge Artificial Intelligence
- ML for Real-World Deployment: What truly matters once a model leaves the notebook
- Edge AI based on FPGA: From trained model to real-time hardware inference

Introduction to Machine Learning and Edge AI

Introduction to Machine Learning and Edge AI

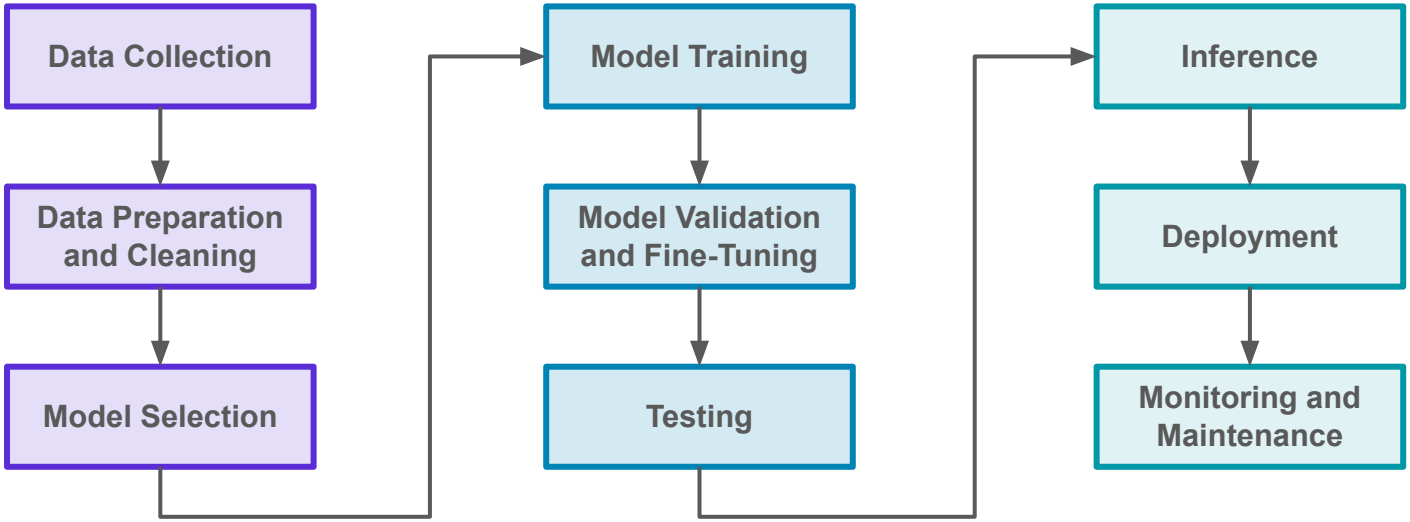


A machine/system capable of imitating human behavior.

Mathematics and statistics to learn from data.

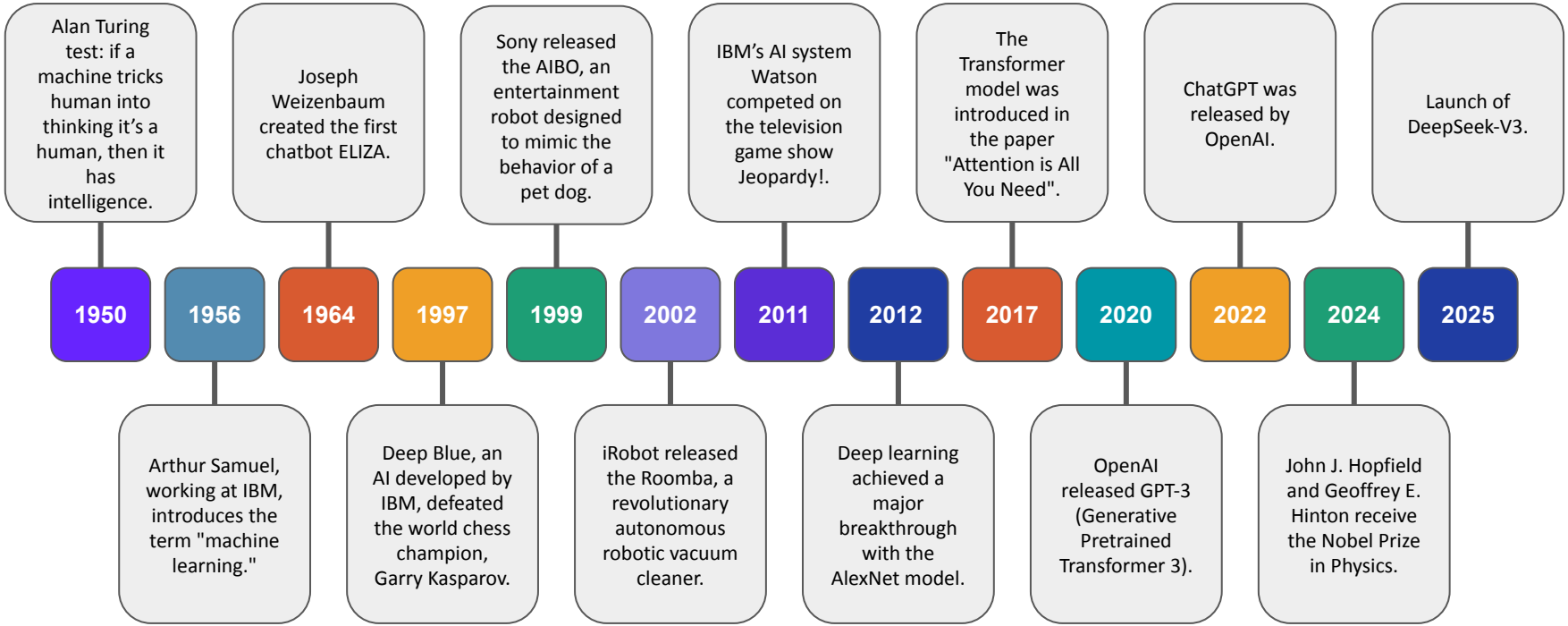
Neural networks multi-layer.

Introduction to Machine Learning and Edge AI



[Based on the original from Qualcomm Academy]

Introduction to Machine Learning and Edge AI



Why has the explosion of AI happened in recent years?

Introduction to Machine Learning and Edge AI

Why has the explosion of AI happened in recent years?

Big data

The massive amount of data generated every day (from social media, sensors, and online activity) provides the fuel that AI systems need to learn and improve

Introduction to Machine Learning and Edge AI

Why has the explosion of AI happened in recent years?

Big data

The massive amount of data generated every day (from social media, sensors, and online activity) provides the fuel that AI systems need to learn and improve

Software

New algorithms, neural network architectures, and open-source frameworks have dramatically improved AI's capabilities and made development more accessible.

Introduction to Machine Learning and Edge AI

Why has the explosion of AI happened in recent years?

Big data

The massive amount of data generated every day (from social media, sensors, and online activity) provides the fuel that AI systems need to learn and improve

Software

New algorithms, neural network architectures, and open-source frameworks have dramatically improved AI's capabilities and made development more accessible.

Hardware

Advances in GPUs, TPUs, and cloud computing have made it possible to train large AI models faster and more efficiently than ever before.

Edge
Artificial Intelligence

Edge AI

“Edge artificial intelligence (Edge AI) refers to the deployment of AI algorithms and AI models directly on local edge devices such as sensors or Internet of Things (IoT) devices, which enables real-time data processing and analysis without constant reliance on cloud infrastructure.” [IBM]

[IBM] https://www.ibm.com/think/topics/edge-ai

Edge AI

“Edge artificial intelligence (Edge AI) refers to the deployment of AI algorithms and AI models directly on local edge devices such as sensors or Internet of Things (IoT) devices, which enables real-time data processing and analysis without constant reliance on cloud infrastructure.” [IBM]

Low Latency

Real-time decisions, reducing cloud dependence

Privacy & security

Data stays on the device.

Scalability & reliability

Devices run autonomously, even offline.

Lower Bandwidth Use

Less data sent to the cloud.

Energy efficiency

Optimized for local processing.

[IBM] https://www.ibm.com/think/topics/edge-ai

Edge AI

“Edge artificial intelligence (Edge AI) refers to the deployment of AI algorithms and AI models directly on local edge devices such as sensors or Internet of Things (IoT) devices, which enables real-time data processing and analysis without constant reliance on cloud infrastructure.” [IBM]

Low Latency	Privacy & security	Scalability & reliability	Lower Bandwidth Use	Energy efficiency
Real-time decisions, reducing cloud dependence	Data stays on the device.	Devices run autonomously, even offline.	Less data sent to the cloud.	Optimized for local processing.

Bringing intelligence closer to the data, enabling low-latency, privacy-preserving, and energy-aware AI.

[IBM] https://www.ibm.com/think/topics/edge-ai

Edge AI

Challenges

Edge AI
Challenges

Hardware & Energy

Limited power

Memory / storage

limits

Thermal constraints

Edge AI
Challenges

Hardware & Energy

Limited power

Memory / storage limits

Thermal constraints

Computation & Performance

Latency & real-time

Limited compute capacity

Hardware heterogeneity

Model size vs accuracy tradeoff

Edge AI
Challenges

Hardware & Energy

Limited power

Memory / storage limits

Thermal constraints

Computation & Performance

Latency & real-time

Limited compute capacity

Hardware heterogeneity

Model size vs accuracy tradeoff

Data & Deployment

Privacy & security

Network issues

Data distribution shift

Edge AI

Heterogeneous Computing: A Key Enabler for Edge AI

CPU

Few cores
Poor parallelism for ML inference

High latency
Slow for compute-intensive tasks

Not optimized for inference
General purpose, not ML-specific

GPU

High power consumption
Not suitable for battery-powered edge

Size and cost
Too large for small embedded systems

Memory bandwidth bottleneck
Data movement limits throughput

FPGA

Hard to program
Requires HDL expertise

Limited flexibility
Not general purpose

Longer development cycle
Synthesis and timing closure

What if we could combine the strengths of different processors CPUs, GPUs, and FPGAs to overcome these limits?

Edge AI

Heterogeneous Computing: A Key Enabler for Edge AI

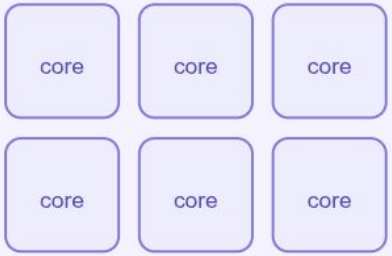
By incorporating CPUs, GPUs, and FPGAs into a single system, **heterogeneous computing** becomes possible.

This approach maximizes the strengths of each component, efficiently distributing edge workloads to improve both performance and energy consumption.

Edge AI

Heterogeneous Computing: A Key Enabler for Edge AI

CPU

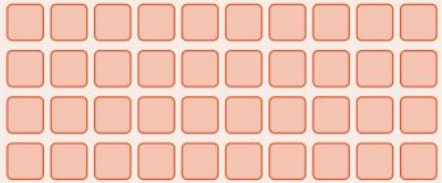


Few powerful cores

Control logic · system mgmt

lightweight AI inference

GPU

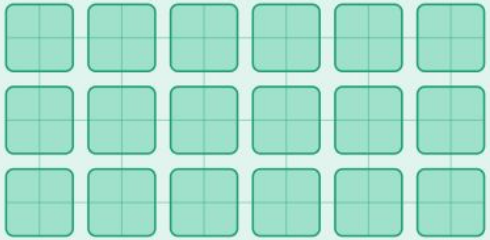


Thousands of small cores

Deep learning inference

Image processing · HPC

FPGA



Reconfigurable logic blocks

Ultra-low latency

Custom AI accelerators · real-time

Edge AI

Current trends

NPU Integration in SoCs

Chip manufacturers are embedding dedicated **Neural Processing Units (NPUs)** directly into their SoCs.

- **Qualcomm Hexagon**
- **Apple Neural Engine**
- **MediaTek APU**

Optimized Inference Frameworks

Leading frameworks optimized for edge inference:

- TensorFlow Lite
- ONNX Runtime
- Core ML
- OpenVINO

TinyML Convergence

The concept is merging with TinyML, bringing AI even to ultra-low-power microcontrollers previously considered too limited.

µW	KB	ms
Power consumption	Memory footprint	Inference latency

Extreme resource constraints met with AI efficiency

Edge AI represents a paradigm shift: from centralized cloud-dependent AI toward distributed, autonomous, and situated intelligence (one of the most important growth vectors in the AI ecosystem).

Edge AI

The Deployment Gap

Problem:

Modern ML models are often large, resulting in high memory consumption and slow inference.

The most accurate models (such as deep neural networks) are large and expensive in terms of memory and processing time.

Edge AI

The Deployment Gap

Model	Parameters	Disk size
BERT-large	~340M	~1.3 GB
VGG-19	~144M	~548 MB
VGG-16	~138M	~528 MB
BERT-base	~110M	~440 MB
YOLOv3	~62M	~236 MB
ResNet-50	~25.6M	~98 MB
SpineNet-49S	~11M	~45 MB
MobileNetV3-Large	**~5.4M**	**~20 MB**

Green = compressed / efficient model target · All others are candidates for compression

Edge AI

The Deployment Gap

Problem:

Modern ML models are often large, resulting in high memory consumption and slow inference.

Solution:

Apply model compression techniques to reduce size and improve efficiency.

Edge AI

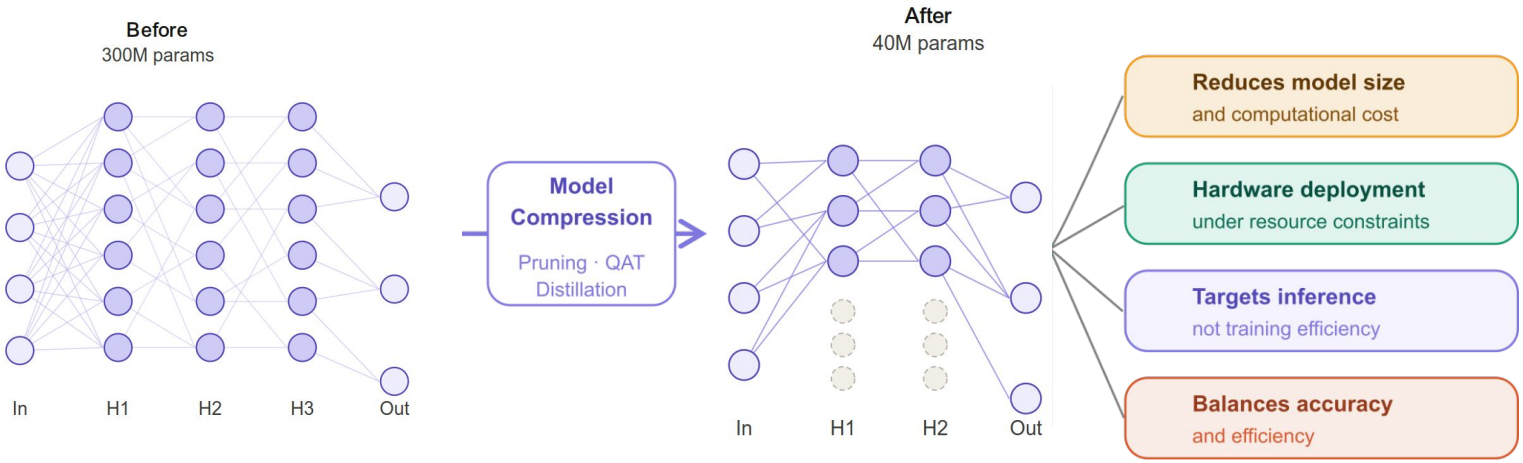
The Deployment Gap

Model	Parameters	Disk size	Accuracy
DistilBERT	66M ↓ 40%	~250 MB	97% of BERT
SqueezeNet 1.2M	1.2M ↓ 99%	~4.8 MB ↓ 99%	Similar to VGG-16
YOLOv8-Nano	3.2M ↓ 92%	~8 MB ↓ 90%	Good trade-off with YOLOv8-L
YOLO-Fastest	0.2M ↓ 99.5%	~1 MB ↓ 99%	Lower accuracy

These examples show that significant reductions in model size are possible with limited impact on accuracy

Edge AI

Compression



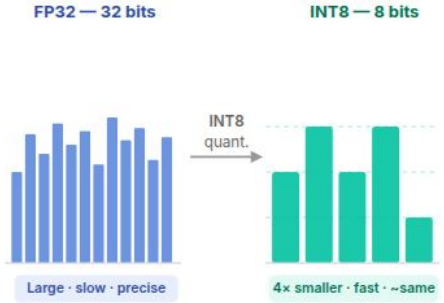
Model compression reduces size and complexity to enable efficient inference under real-world constraints

Edge AI

Compression

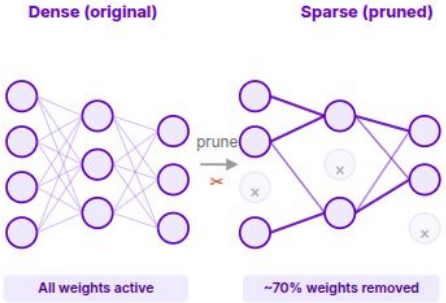
Quantization

Reduce weight precision to shrink model size and speed up inference



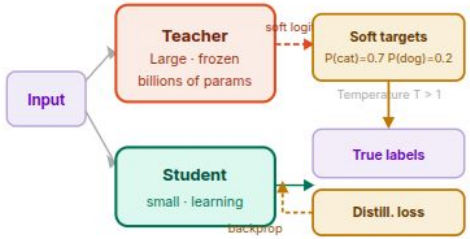
Pruning

Remove low-importance connections to create a sparse, leaner network



Knowledge Distillation

A large teacher model trains a compact student model via soft targets



Edge AI

Compression

Model compression is not a single technique,
but a design space where accuracy, efficiency, and hardware constraints must be balanced.

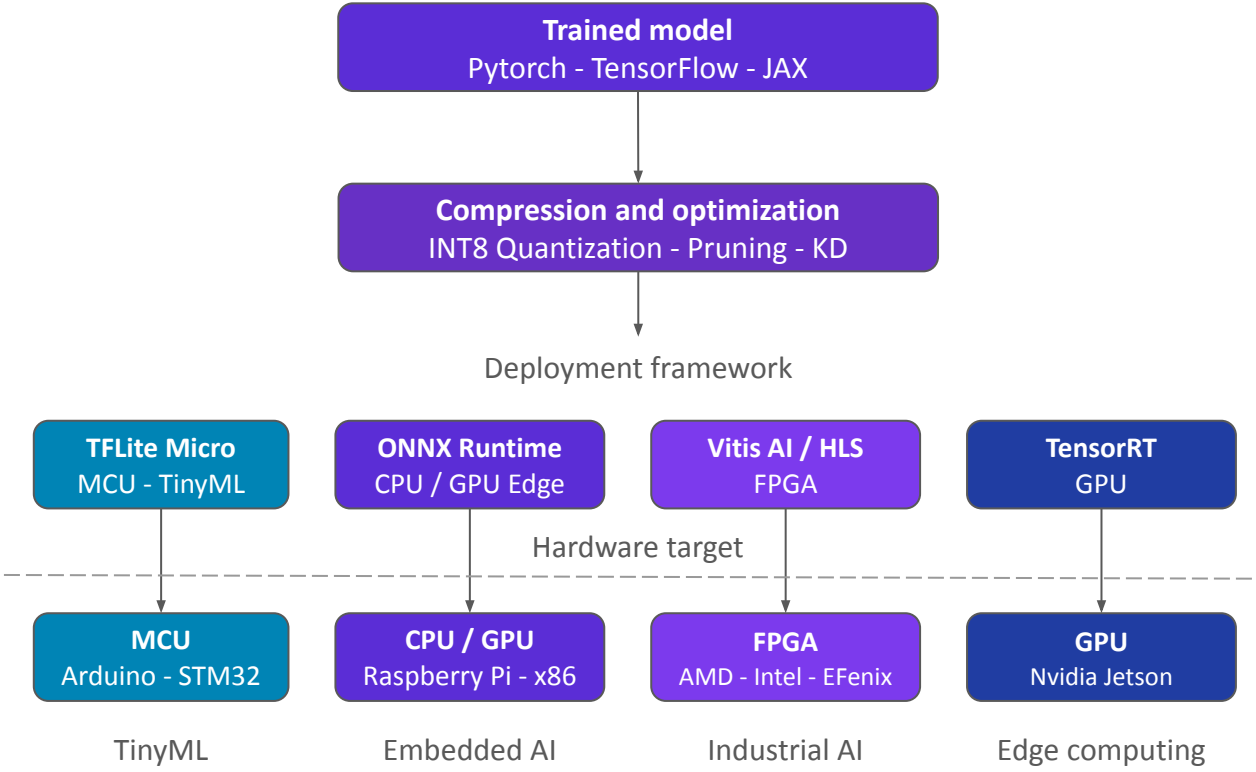
Edge AI

Compression

Model compression is not a single technique,
but a design space where performance metrics, efficiency, and hardware constraints must be balanced.

Every optimization (pruning, quantization, or distillation) improves efficiency at the cost of something else. The goal is not to avoid trade-offs, but to choose the right ones.

Edge AI
Ecosystem



ML for Real-World Deployment

What truly matters once a model leaves the notebook

Research vs Deployment

Research

The question:

Does it reach 98% accuracy on the test set?

Success metric:
accuracy, F1, AUC

Research vs Deployment

Research

The question:

Does it reach 98% accuracy on the test set?

Success metric:
accuracy, F1, AUC

vs

Deployment

The question:

Does it run in 10ms with 64 KB of RAM?

Success metric:
latency, memory, energy, accuracy within tolerance

The four deployment constraints

Latency

< 10ms real-time
< 100ms near real-time
< 1s batch acceptable

Memory

MCU: 64–256 KB flash
FPGA: 2–8 MB BRAM
Edge GPU: 256 MB+

Energy

Battery: µW–mW budget
Mains: less critical
Harvesting: µW limit

Accuracy tolerance

Safety-critical: > 99%
Industrial: > 95%
Consumer: > 90%

Conventional AI vs Edge AI

Conventional AI
Cloud/server-based inference

High-latency
Network round-trips add delay

High energy demand
TOPS-scale compute, not battery-friendly

Privacy exposure
Raw data sent to external servers

Deep neural networks
Billions of params, dense matrix operations

Ineffective at the extreme edge
Alternative paradigms needed

Conventional AI vs Edge AI

Conventional AI Cloud/server-based inference	Edge AI On-device, local processing
High-latency Network round-trips add delay	**Real-time response** Minimal latency, no network round-trip
High energy demand TOPS-scale compute, not battery-friendly	**Ultra-low power consumption** uW to mW range, long battery life
Privacy exposure Raw data sent to external servers	**Privacy-aware computation** Data never leaves the device
Deep neural networks Billions of params, dense matrix operations	**Dynamic multisensory input** Different sensors, processed locally
Ineffective at the extreme edge Alternative paradigms needed	**Efficient local intelligence** New paradigms required

Optimizing every phase of the design and implementation process

Optimizing every phase of the design and implementation process

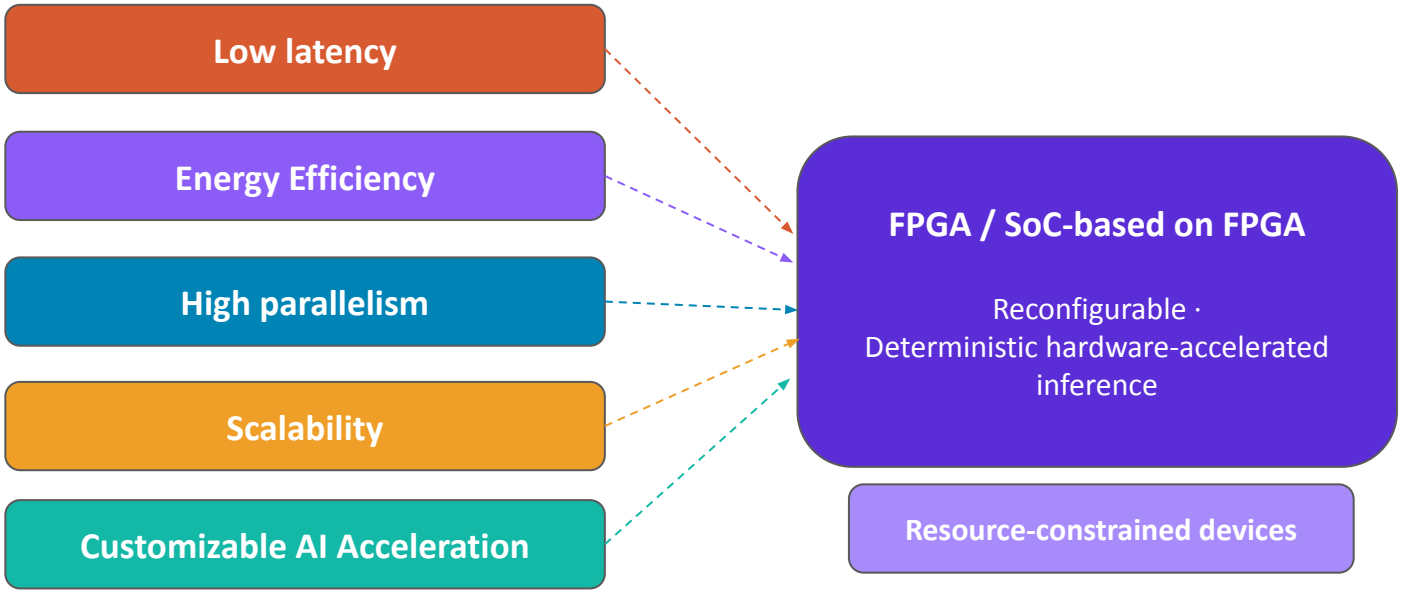
Software development	HW platform selection	Firmware creation	PCB design
Model training	Edge devices	C/C++ application	Key components
Model compression	Cloud	Python application	Signal and power integrity
	GPU/CPU	HDL	Methodological design
	Others		EMC and EMI

Edge AI based on FPGA

From trained model to real-time hardware inference

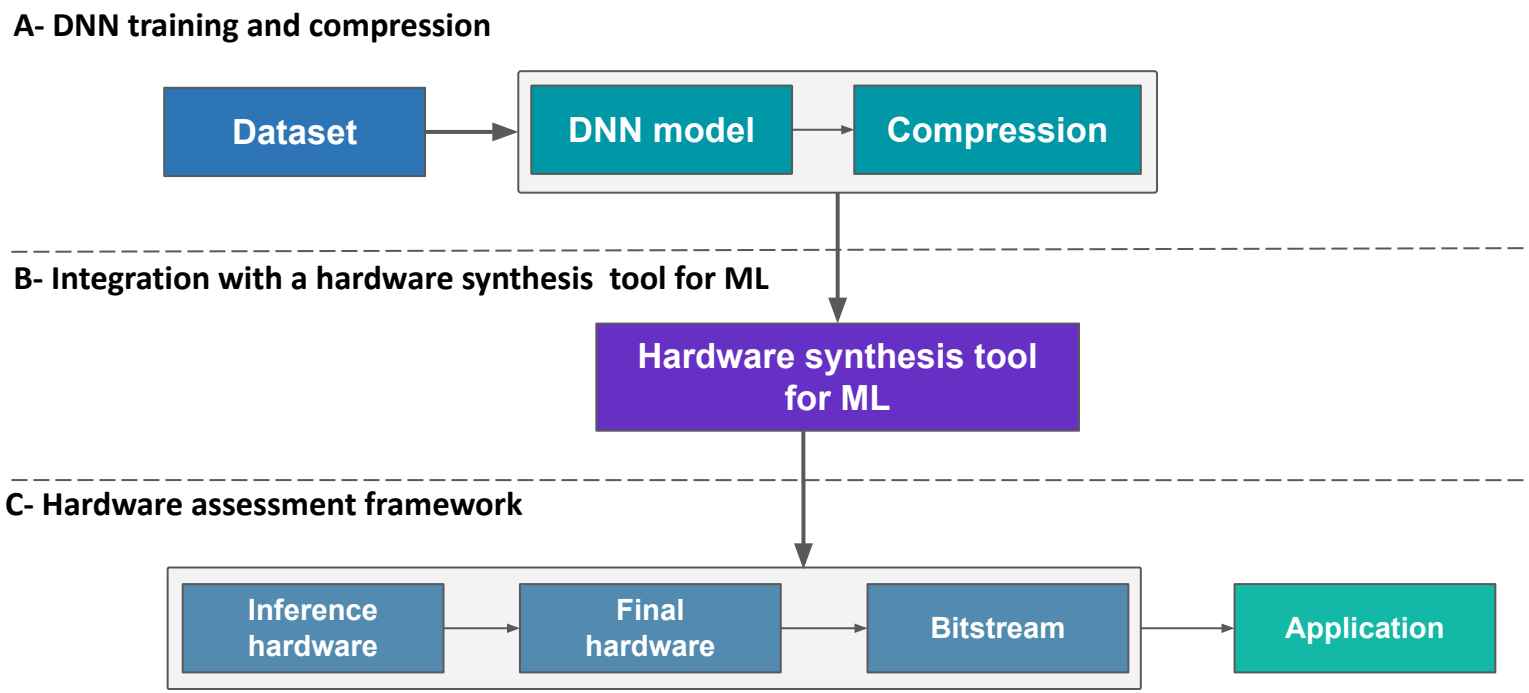
Edge AI based on FPGA

From trained model to real-time hardware inference



Edge AI based on FPGA

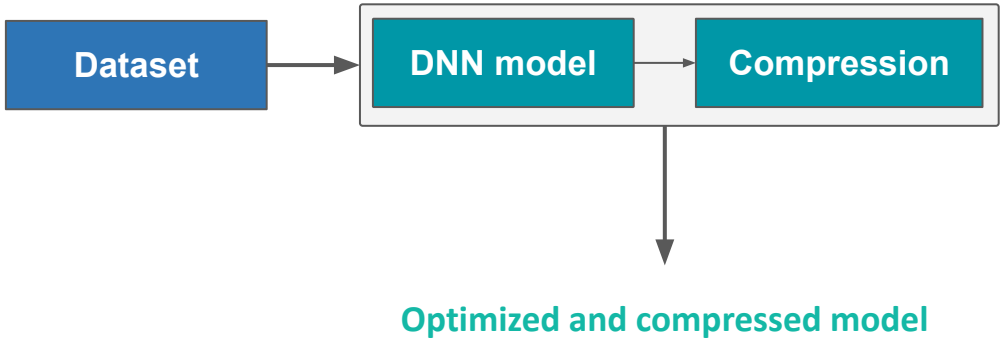
From trained model to real-time hardware inference



Available at https://github.com/RomiSolMolina/workflowCompressionML

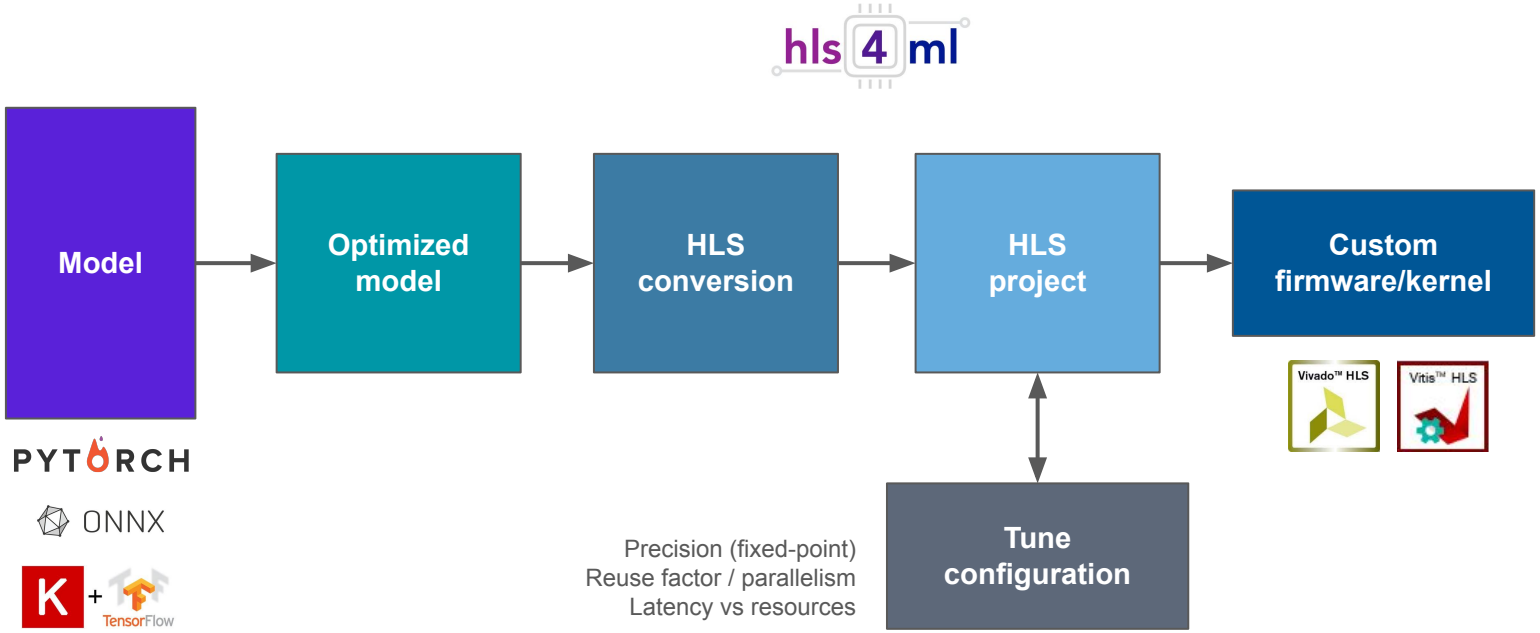
Edge AI based on FPGA

From trained model to real-time hardware inference



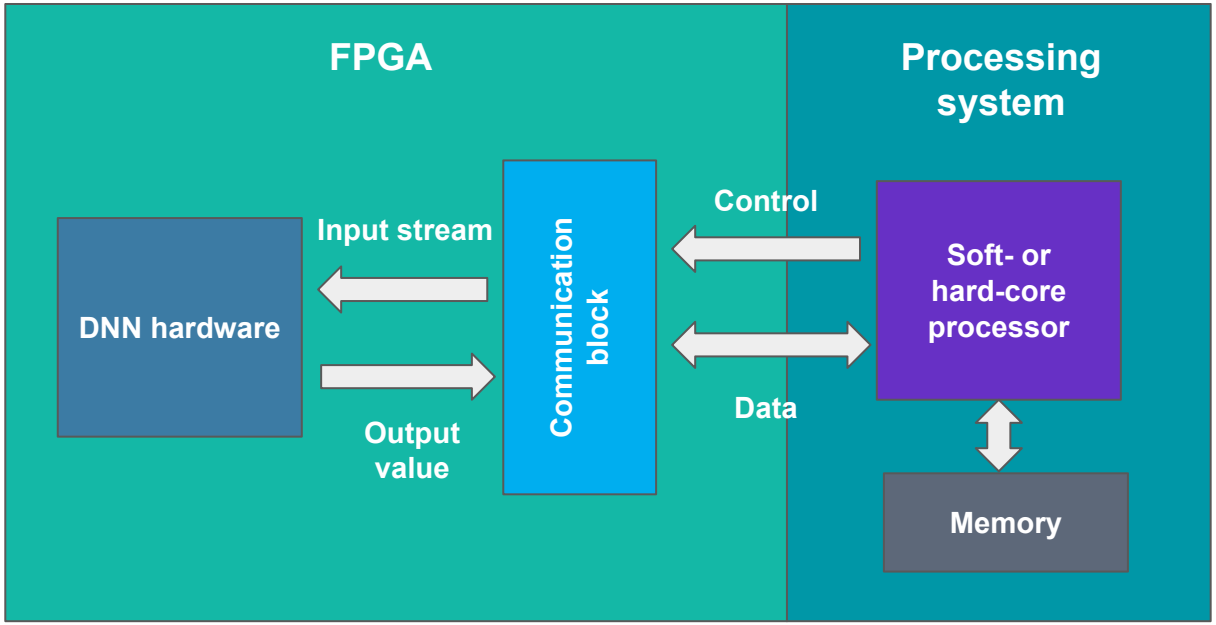
Edge AI based on FPGA

From trained model to real-time hardware inference



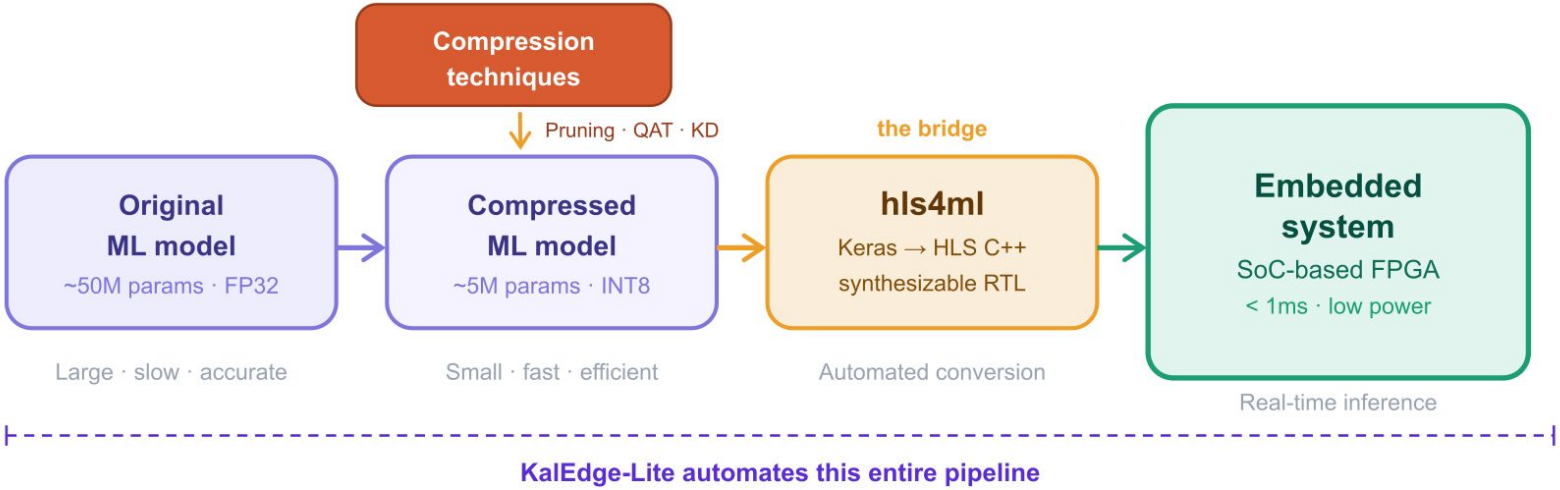
Edge AI based on FPGA

From trained model to real-time hardware inference



Edge AI based on FPGA

From trained model to real-time hardware inference



Thank you!

Romina Soledad Molina, Ph.D.

rsmolina@kaleidoforge.com

KaleidoForge
Lightning Bridges for Fast AI

UCLM
Universidad
Castilla-La Mancha

Edge AI and Machine Learning

Romina Soledad Molina, Ph.D.

Ciudad Real, Castilla-La Mancha, España - 2026

KaleidoForge
Lightning Bridges for Fast AI

UCLM
Universidad
Castilla-La Mancha